

Date: 01.10.26

* =SUM() : It adds the range of cells. that adds the total range of cells and do the addition.

Syntax : =SUM(num1)

Ex : =SUM(C2:C21)

Output : ₹ 1,84,600.00 → Total basic salary

* =AVERAGE() : It returns the Average value of the selected data.

Syntax : =AVERAGE(num1)

Ex : =AVERAGE(C2:C21)

Output : ₹ 9,230.00

* =MAX() : It will returns the maximum or the largest value in the selected cells.

Syntax : =MAX(num1)

Ex : =MAX(C2:C21)

Output : ₹ 16,250.00

* =MIN() : It will returns the minimum or the smallest value in the selected cells.

Syntax : =MIN(Num1)

Ex : =MIN(C2:C21)

Output : ₹ 4,500.00

* =COUNT() : It counts the Number of cells with Numbers having the data: that will counts.

Syntax : = COUNT (Value 1)

Ex : = COUNT (C2:C21)

Output : 20

In the same way we can calculate or Count the Number of employees

Ex : = COUNT (B2:B21)

Output : 20

* LARGE() : This Function will select the particular array that we want like 5th highest (or) 2nd highest Number in the Selected range.

Syntax : = LARGE (array, K) K = ~~2nd~~ 5th largest

Ex : = LARGE (C2:C21, 5)

Output : 11250

* SMALL() : This Function will select the particular array that we want from 10th smallest number in the Selected range.

Syntax : = SMALL (array, K)

Ex : = SMALL (C2:C21, 10)

Output : 8750

* LARGE + UNIQUE() : This function is same as the large function. But UNIQUE is function that a given largest without counting duplicates it will remove the duplicate.

Syntax : = LARGE (UNIQUE (array, K)

Ex : = LARGE (UNIQUE (C2:C21), 2)

Output : 12000

* =small + UNIQUE(): This is same as the small function. But UNIQUE removes the duplicate.

Syntax: =small(UNIQUE(array), k)

Ex: =small(UNIQUE(C2:C21), 3)

Output: 6275

* =SUMIF(): This function helps us to find the total salary of marketing team in the given list of a department.

Syntax: =SUMIF(range, Criteria, [sum-range])

Ex: =SUMIF(F2:F21, F4(OR)marketing,

Output: 31500 C2:C21)

* In the same case sumif function we can use like find total salary of employees who are all earning 10000.

Syntax: =sumif(C2:C21, 10000)

Output: 30000

we can also find the total salary of a employee who are all earning above 10000.

Ex: =sumif(C2:C21, ">10000")

is a string due to that

Output: 90250

* To Find Average salary of a west region employee in the department

Syntax: =Average(range, Criteria, [average-range])

Ex: $=AVERAGE(F2:F21, E2, G2:G21)$

Output: ₹ 8500.00

* To Find the maximum salary in r & d department
=MAXIFS it finds the maximum (highest) value that satisfies a condition.

Syntax: $=MAXIFS(\text{max-range}, \text{criteria-range1}, [\text{criteria2}])$

Ex: $=MAXIFS(G2:G21, F2:F21, F5)$

Output: 10000

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* =SUMIFS: This function used to satisfy the more conditions in the statement.

→ Find total salary of marketing team of north region employee

Syntax: $=SUMIFS(\text{sum-range}, \text{criteria-range1}, \text{criteria2})$

Ex: $=SUMIFS(G2:G21, F2:F21, F8, E2:E21, "N")$

Outputs: 15750

* =AVERAGEIFS: Calculates the average (arithmetic mean) of a range of numbers based on multiple considerations or criteria.

→ Calculate the Average Salary of the employee who are earning more than 10K Salary in the training department.

Syntax: $=AVERAGEIFS(\text{Avg-range}, \text{criteria range1}, \text{criteria2})$

Ex: $=AVERAGEIFS(G2:G21, G2:G2, 10000, F2:F21, F21)$

Output: 10000

* =COUNTIFS: it counts the number of cells in a range that meets more than a one condition.

→ How many employees working from Jayanagar R&D department in South region.

Syntax: =COUNTIFS(criteria_range1, criteria1, ...)

Ex: =COUNTIFS(C2:C21, C2, D2:F21, F13, E2:E21, E17)

Output: 0

→ To Calculate the DA 8% of Basic then just multiply with 8% with Basic

Ex: $EX = C2 * 8\%$ $C2 = (\text{Basic}) ₹10000.00$
 $\times (=C2 * 0.008) \times$

EX = $C2 * 0.08$

Output: ₹800.00

→ TA of 4000 BONUS TO Marketing team of the department. We should lock the Marketing team

Ex: =IF(F2 < \$F\$4, 4000, 0)

Output: Only Marketing team members can get the "4000" Bonus

Syntax: IF(logical_test, [value_if_true], [value_if_false])

→ HRA 12% of Basic and DA.
By adding the Both Basic and DA and Multiply
With 12% HRA.

$$\text{Ex} : = (C2 + D2) * 12\%$$

Output : ₹ 1,296.00

→ BONUS of 3000 Accounts Department South region
Employee

Syntax : $=IF(AND(\text{logical test}, [\text{value if true}], [\text{value if false}]))$
Ex : $=IF(AND(F2=\$F\$3, E2=\$E\$5), 3000, 0)$

Output : 0 if Non-Account, south region
3000 if Account, south region.

→ Incentive 5000 to Bangalore or operation Department
We should lock the Bangalore and the Operation
Department According to the given two con-
ditions.

Syntax : $=IF(OR(\text{logical test}, [\text{value if true}], [\text{value if false}]))$

$$\text{Ex} : =IF(OR(D2=\$D\$11, F2=\$F\$18), 5000, 0)$$

Output : 0 if Non Bangalore or Operation department
5000 if Bangalore or Operation department

→ Provident Fund 15% of Basic
Should Multiply the Basic with PF 15%.

$$\text{Ex} : =C2 * 15\%$$

Output : ₹ 1,500.00

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* IFS(): This function will satisfy the more than two conditions.

→ Nested if(): It places one if statement inside another to test multiple conditions and return different outcomes.

Tax Ex: $\text{IF}(C2 \leq 5000, 0, \text{IF}(C2 \leq 10000, 100, \text{IF}(C2 \leq 15000, 200, \text{IF}(C2 \leq 20000, 300, 400))))$

Output: According to the given conditions.

→ Tax Ex: $\text{IFS}(C2 \leq 5000, 0, C2 \leq 10000, 100, C2 \leq 15000, 200, C2 \leq 20000, 300, C2 \geq 20000, 400)$

→ To Calculate the Net salary. We have to add up the Basic + DA + HRA + TA + Bonus + Incentive.

$$\text{Ex: } = C2 + H2 + I2 + J2 + K2 + L2$$

Output: ₹ 12,096.00

→ To Calculate the Gross salary, we should add up the Net salary + PF + TAX

$$\text{Ex: } = T2 + M2 + N2$$

Output: ₹ 13,696.00

FINANCIAL FUNCTIONS :-

* Principal Amt. Rate of Interest Time Future value
 ↓ ↓ ↓
 65000 9.00% 5yr ₹1,00,010.56
 is due to EMI, Bank cuts Amt.

Syntax: $=FV(\text{rate}, \text{nper}, \text{pmt}, [\text{pv}], [\text{type}])$

↓
 it's for monthly

→ rate = interest
 nper = duration / Time
 PV = Principal Value

* 1. Mr. Sunil purchased a Bike for Rs. 65000 by taking a loan from Bank, repayable in 3 years by monthly installments with interest of 8.5% pa. Calculate the EMI for the loan [use PMT]

PV = 65000, Rate = 8.50%, Time = 3yr, PMT = ?

Syntax: $\text{PMT}(\text{rate}, \text{nper}, \text{PV}, [\text{FV}], [\text{type}])$

Ex: $\text{PMT}(C13/12, D13*12, B13)$

Output: ₹ 2,051.89 → PMT

* 2. Sunil wants to know the breakup of the principal amount in the EMI payable for the 1st month [PPMT]

PV = 65000, Rate = 8.50%, Time = 3, PPMT = ?

Syntax: $\text{PPMT}(\text{rate}, \text{per}, \text{nper}, \text{PV}, [\text{FV}], [\text{type}])$

Ex: $\text{PPMT}(C16/12, 1, D16*12, B16)$

Output: ₹ -1,591.47 → PPMT

* 3. Now Sunil wants to know the Breakup of the amount in the EMI Payable for the 1st interest month. [IPMT]

PV = 65000, Rate = 8.50%, Time = 3, IPMT = ?

Syntax : IPMT (rate, per, nper, PV, [FV], [type])

Ex : =IPMT(C19/12, 1, D19*12, B19)

Output : ₹ - 460.42 → IPMT

* 4. Meena has a fixed deposit of Rs. 25000/- in Bank for 5 yrs. on which interest given by the bank is 8.25%. Calculate the maturity value of Meena [FV] → future value

PV = -25000, Rate = 8.25%, Time = 5, FV = ?

Syntax : =FV (rate, nper, pmt, [PV], [type])

Ex : =FV(C22, D22, B22)

↓
pmt No. value

Output : ₹ 37,160.33 → FV

* 5. Rita is having to option of going for RD in which amount payable per month is Rs. 12000, interest is 7.5% and the period of deposit will be 5 yrs. She asked you to calculate the maturity value for this option [FV]

RD = -1200, Rate = 7.50%, Time = 5, FV = ?
(B) (C) (D)

Syntax: $FV(\text{rate}, \text{nper}, \text{pmt}, [\text{pv}], [\text{type}])$

Ex: $=FV(C25/12, D25*12, B25)$

Output: ₹ 87,032.53

*6. JP wants to invest in a plan where he will pay months is Rs. 1200, interest is 7.5% and the period of deposit will be 5 yrs - she a) X

*6. JP wants to invest in a plan where he will pay Rs. 1476 quarterly for 10 years to get a maturity value of Rs. 80,000. He wants you to calculate the rate of interest he will earn on this investment [RATE].
PV = -1476, Time = 10, FV = 80000, Rate = ?

Syntax: $RATE(\text{nper}, \text{pmt}, \text{pv}, [\text{fv}], [\text{type}], [\text{guess}])$
Ex: $=RATE(C28*4, B28, D28)$

Output: 10.5% → Rate

They mention quarterly. So we have to multiply the Time in to 4

Rate = 10.5%

*7. Before opting for the plan with the rate calculated above and for the same period & maturity value. JP now wants to know the present value of investment Plan [PV].

Time = 10, FV = 80000, Rate = 10.80%, PV = ?

Syntax: PV(rate, nper, pmt, [fv], [type])

Ex : =PV(D31, B31*4, , C31)

They want same period of time quarterly

Output : ₹ 1,476.05 → PV

* 8. Bini wants to go for a RD scheme in any Bank with an amount of Rs. 1520/- Per month where the interest is payable is 7.5% per annum. She needs your help in finding the no. of installments she has to pay (also cal. the no. of years) to get a maturity

Value of Rs. 50000 [NPER]

RD = 1520, Rate = 7.5%, FV = 50000, NPER = ?

Syntax: NPER(rate, pmt, PV, [FV], [type])

Ex : =NPER(E34/12, B34, , D34)

O/p = 30-months (NPER)

to Calculate months to year

Ex : =E34/12

O/p : 2.500726627